# Software Requirements Specification for Flexin: subtitle describing software

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# Revision History

| Date | Version | Notes |
|---|---|---|
| Date 1 | 1.0 | Notes |
| Date 2 | 1.1 | Notes |

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*Insert your content here.*

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## 15 Security Requirements

### 15.1 Access Requirements

| ID | Requirement | Specification |
|---|---|---|
| SEC-ACC-01 | Authentication and Sessions | **Description:** Users shall authenticate before accessing personal fitness data, portfolios, or transactions. The React Native client shall authenticate with the Spring API using secure token-based sessions.<br>**Rationale:** Authentication protects personal data and virtual assets from unauthorized access.<br>**Fit Criterion:** All protected API endpoints shall reject missing or invalid access tokens with HTTP 401. Access tokens shall expire within 60 minutes, with refresh or re-authentication required. |
| SEC-ACC-02 | Role-Based Authorization | **Description:** Role-based access control shall restrict users to their own records and limit administrative or evaluator access to authorized functions.<br>**Rationale:** Access restrictions prevent unauthorized modifications and cross-account data exposure.<br>**Fit Criterion:** JUnit security tests shall confirm that attempts by standard users to access or modify another user's records return HTTP 403 in all tested cases. |
| SEC-ACC-03 | Fitness Integration Authorization | **Description:** Flexin shall request only necessary read permissions for supported integrations, Android Health Connect and Garmin. Integration credentials shall be stored using secure, Android Keystore-backed storage.<br>**Rationale:** Secure credential handling protects external accounts and limits access to fitness data.<br>**Fit Criterion:** Integration tokens shall use Android Keystore-backed secure storage. No credentials, tokens, or secrets shall be committed to the project repository. |

### 15.2 Integrity Requirements

*Insert your content here.*

### 15.3 Privacy Requirements

*Insert your content here.*

### 15.4 Audit Requirements

*Insert your content here.*

### 15.5 Immunity Requirements

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# 25 Waiting Room

*Insert your content here.*

# 26 Ideas for Solution

*Insert your content here.*

# 27 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

## 27.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

## 27.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

## 27.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

# 28 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?